

Reviewer Pack — Minimal Geometric Entropy of Affine Schemes Bounds Resolution...

Entry c60ff81cd1ee · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Geometric Entropy of Affine Schemes Bounds Resolution Proof Size

Entry ID: c60ff81cd1ee **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-30 10:22:52 UTC

1. Conjecture

Field A (mathematical branch): Affine Geometry **Field B** (complexity object): Boolean Function Complexity: DPLL Refutation Complexity

Statement:

For any Boolean function f with n variables, let A_f be the associated affine scheme defined by the polynomial of f . The geometric entropy of A_f , denoted as $GE(A_f)$, is upper bounded by a constant multiple of the DPLL refutation size $t(f)$. *Specifically, $GE(A_f) = \Theta(t(f))$.*

Rationale (proposer's reasoning):

Affine geometry provides a geometric interpretation of Boolean functions, which could potentially reveal underlying structure related to computational complexity. The geometric entropy captures the complexity of the scheme and may correlate with the proof length required for DPLL refutation.

Taxonomy category: cg_kw_andreev (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 9c0a48c107368435

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all Boolean functions f with n variables ($n \leq 40$), the geometric entropy $GE(A_f)$ divided by the DPLL refutation size $t^*(f)$ is less than or equal to a constant factor k , and no seed produces a ratio greater than $k + 1$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.90	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "geometric entropy" AND "affine scheme" AND "Boolean function complexity" - "DPLL refutation complexity" IN AFFINE geometry" - "polynomial of Boolean functions" AND "upper bound" AND "geometric entropy"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.3s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.randint(0, 1) for _ in range(2**n)]

    def dpll_refutation_size(f):
        n = len(f)
        clauses = []
        for i in range(n):
            clause = []
            for j in range(i + 1, n):
                if f[2**(i + j)] != f[2**i] ^ f[2**j]:
                    clause.append((i, j))
            clauses.append(clause)
        return len(clauses) + n

    def geometric_entropy(f):
        n = len(f)
        count = [0] * (n + 1)
        for i in range(2**n):
            count[bin(i).count('1')] += 1
        entropy = 0
        total = 2**n
        for c in count:
            if c > 0:
                p = c / total
                entropy -= p * math.log2(p)
        return entropy
```

```

n_max = 40
instances_tested = 30
metric_values = []

for _ in range(instances_tested):
    n = random.randint(5, n_max)
    f = generate_boolean_function(n)
    ge = geometric_entropy(f)
    t_f = dpll_refutation_size(f)
    if t_f > 0:
        ratio = ge / t_f
        metric_values.append(ratio)

return {
    "metric_name": "GE(A_f) / t*(f)",
    "metric_value": sum(metric_values) / len(metric_values),
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": all(x <= 2 for x in metric_values), # Assuming k = 1
    "counterexample": "" if all(x <= 2 for x in metric_values) else "GE(A_f) / t*(f) > 2"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [random.randint(1, 1000) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"GE(A_f) / t*(f) > 2\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means the pre-registered support condition could not be unambiguously met. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	29302
2	propose	ollama_remote	glm4:latest	0	0	23696
3	propose	ollama_remote	glm4:latest	0	0	14081
4	preregistration	ollama_remote	glm4:latest	0	0	12597
5	novelty	ollama_remote	glm4:latest	0	0	11436
6	novelty	ollama_remote	glm4:latest	0	0	16250
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11841
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	42860
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15219
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9496
11	judge	ollama_remote	glm4:latest	0	0	26218

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 212995 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in research/audit/c60ff81cd1ee.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/c60ff81cd1ee.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/c60ff81cd1ee.tar.gz (if generated)